Sonu Yadav

Senior Research Scientist (Researcher III)
Department of Physics and Astronomy,
West Virginia University

Contact Information	Personal Details
Department of Physics and Astronomy West Virginia University, Morgantown West Virginia, USA Phone: (M) +1 681 285 0380 Email: sonu.yadav@mail.wvu.edu sohamyadav1908@gmail.com	Sex: Male Marital Status: Married, Nationality: Indian Language: English, Hindi

Career Highlights

I am an experimental plasma physicist with more than ten years of experience in plasma science, specializing in magnetized plasmas, plasma–surface interactions, and advanced diagnostic development. My research addresses problems of national importance across space weather prediction, semiconductor manufacturing, and space propulsion plasma systems. My technical expertise includes magnetic reconnection, MHD-to-kinetic-scale turbulence, wave–particle interactions, plasma instabilities, and precision plasma etching and deposition, supported by extensive experience in plasma source development, computer-controlled experimental operations, and kinetic modeling.

I earned my Ph.D. from the Institute for Plasma Research (IPR), India, and subsequently completed 5 years of postdoctoral research at West Virginia University (WVU) and IPR. I currently serve as a Senior Research Scientist at WVU, where I lead experimental operations on the PHAse Space MApping (PHASMA) facility, a nationally unique user platform for studying collisionless magnetic reconnection, MHD-to-kinetic-scale plasma turbulence and energy dissipation, and particle energization relevant to Earth's magnetosphere and space weather. In this role, I independently direct complex experimental campaigns, mentor Ph.D. researchers, and develop new high-resolution diagnostics for electron and ion velocity distribution measurements. In parallel, I lead an internally funded independent research program on atomic layer etching of two-dimensional materials, applying fundamental plasma kinetics to atomic-scale semiconductor fabrication.

My work integrates RF helicon and inductively coupled plasma source optimization, laser-based diagnostics (Thomson scattering and laser-induced fluorescence), probe-based measurements, and first-principles data analysis to achieve reproducible control of plasma parameters and energy distributions. This research has produced experimental capabilities and scientific insights that directly advance leadership in space science, semiconductor processing, and plasma-based energy technologies. Through sustained independent contributions, mentorship, and cross-disciplinary innovation, I have established myself as a researcher whose continued work provides substantial benefit to the plasma science community.

In addition to my scientific research, I bring a strong cross-functional engineering background directly applicable to semiconductor capital equipment and advanced plasma processing systems:

- **Vacuum Systems**: Extensive experience with UHV and high-vacuum chambers, vacuum feedthrough integration, leak diagnostics, and ICP hardware maintenance
- **Mechanical Integration**: Proficient in SolidWorks with hands-on experience assembling diagnostics, probes, and in situ measurement tools for complex plasma environments
- **Electronics**: Skilled in PCB design (KiCad), circuit simulation (LTSpice), and low-noise signal acquisition for high-precision plasma diagnostics
- **Process Modeling & Data Analysis and Automation**: Fluent in MATLAB, Python, LabVIEW, and LabWindows; experienced in design-of-experiments (DOE), sensitivity analysis, and statistical modeling for process optimization
- **Physics-Based & AI-Assisted Modeling:** Utilize AI-driven tools and first-principles models to interpret experimentally observed plasma physics phenomena and guide experimental design

I am an active contributor to peer-reviewed publications, collaborative research initiatives, and funded proposal development, with a sustained goal of bridging fundamental plasma physics and practical etch process engineering to enable atomic-scale control, plasma-based nanofabrication, and innovation in semiconductor manufacturing.

Research Interests

- **Space weather relevant**: Plasma turbulence, instabilities and transport, Magnetic reconnection, dynamics of plasma flux ropes, and wave–particle interactions
- **Plasma Processing**: Dry Etching; Reactive Ion Etching (RIE), Inductively coupled plasma (ICP)**, and** capacitively coupled plasma (CCP) systems, atomic layer etching (ALE)**,** surface modification**,** and nanoscale fabrication.
- **Electric Propulsion & Expanding Plasma Flows**: Plasma based Thruster, Development and characterization of helicon plasma thrusters**,** magnetized plasma expansion**,** and magnetic nozzle physics, Thrust measurement diagnostics, Mach probe, Laser displacement sensors
- **Plasma Diagnostics & Instrumentation**: Design and deployment of laser-based and in situ diagnostics including Thomson scattering**,** laser-induced fluorescence (LIF)**,** optical emission spectroscopy**,** and probe techniques **(**Langmuir**,** Mach**,** magnetic/B-dot probes**,**

Faraday cup, retarding field energy analyzers (RFEA)) for measuring electron temperature, ion energy, and plasma uniformity in etch-relevant regimes.

Awards and funding:

- **Award type: WVU Internal funding award**
  Title: Seed funding for Plasma Atomic Layer Etching (ALE) of Molybdenum disulphide (MoS2) for next generation semiconductor devices
  PI: Sonu Yadav
  Funding status: Approved (2025)

- **Award type: U.S. department of Energy | office of science (DE-FOA-0003600)**
  Title: Experimental Investigation of Multiple X-Points During Electron-Only Reconnection
  PI: Sonu Yadav (if approved)
  Funding status: Under Review (full submission encouraged)

Honoring:

- **Chair:** Session Chair American Physical Society Division of Plasma Physics 2025
  Session name: Fundamental Plasma: Waves, Instabilities, and Shocks

Postdoctoral Experience

- **Postdoctoral Researcher,** Department of Physics and Astronomy, West Virginia University – May 2022 – Nov 2025
  Conducting laboratory experiments on magnetic reconnection (PHAse Space MApping – PHASMA facility) and space-relevant plasma dynamics. Responsible for operating and maintaining the PHASMA device, including pulsed plasma guns (flux rope sources). Arrays of electric and magnetic probes, and a Thomson scattering diagnostic system to measure electron velocity distribution. Investigating particle heating and turbulence in guide-field reconnection, analogous to plasma processes in space environments. Additionally, developed and tested capacitively coupled plasma (CCP) sources using inert and reactive gases (e.g., chlorine), further extending relevance to semiconductor etching processes. Additionally, I have contributed to the collaborative design, operation, and optimization of capacitively coupled plasma (CCP) sources using both inert (e.g., argon) and reactive (e.g., chlorine) gases, extending the relevance of my work to semiconductor-grade plasma etching and surface processing applications.

- **Postdoctoral Fellow,** *Institute for Plasma Research (Homi Bhabha National Institute), Gandhinagar, India,* May 2020 – May 2022

  Led the development of a 5 kW RF helicon plasma thruster for electric propulsion applications, achieving targeted thrust levels (~100 mN). Designed and operated magnetized expanding plasma columns, magnetic nozzles, and ICP-based plasma sources for simulating space propulsion environments. Acquired extensive experience with high-power RF systems, vacuum-compatible hardware, and plasma–surface interactions.

Skills

Experimental:

Plasma Source Operation & Experimental Systems

- **Plasma Generation Systems:** Extensive hands-on experience with a wide range of discharges including Pulsed RF helicon sources, inductively coupled plasma (ICP) with variable frequencies, capacitively coupled plasma (CCP), pulsed arc plasma guns, ion beam systems, and hot cathode DC discharges—covering pressure regimes and gas chemistries directly relevant to etching, surface modification, and electric propulsion.
- **Etch-Relevant Source Development**: Designed and operated ICP and CCP systems using inert and reactive gases (e.g., Ar, O_2, Cl_2), with emphasis on plasma uniformity, ion energy control, and process scalability for semiconductor fabrication environments.

Plasma Diagnostics & Metrology

- **Probe-Based Diagnostics**: Expert in the development, calibration, and deployment of Langmuir probes (single, double, triple, and RF-compensated), Mach probes (flow measurements), B-dot magnetic probes, Faraday cups, and retarding field energy analyzers (RFEA)—critical for characterizing plasma density, potential, and ion energy distributions in etching conditions.
- **Laser and Optical Diagnostics**: Experienced in Thomson scattering (electron temperature and velocity distributions), laser-induced fluorescence (LIF) (ion velocity distribution functions), and optical emission spectroscopy for real-time monitoring and plasma process optimization.
- **Dynamic Measurement Techniques**: Utilized high-speed imaging, fast camera, fast photodiodes, and synchronized diagnostics for measuring transient plasma events and fluctuations in pulsed and RF-driven systems.

Instrumentation, Integration & Vacuum Systems

- **Vacuum Systems**: Proficient in operating and maintaining UHV and high-vacuum systems**,** vacuum feedthroughs, and custom plasma chambers for high-reliability operation in research and etch-relevant settings.
- **Instrumentation Integration**: Designed and implemented motorized 1D/2D probe drives and in situ diagnostics for spatial plasma profiling. Skilled in RF matching network design**,** grounding/shielding strategies**,** and noise mitigation in sensitive plasma measurements.

Computational & Software:

Data Analysis & Process Optimization

- **Programming & Modeling**: Proficient in MATLAB and Python for scientific computing, diagnostics modeling, complex data analysis, and process optimization. Familiar with Mathematica**,** Origin**,** and basic experience with IDL and C/C++ for data visualization and signal processing.
- **Statistical Tools**: Skilled in Design of Experiments (DOE) and use of COMSOL multi-physics tool for sensitivity analysis and plasma process tuning.
- **Modeling**: Utilize AI-driven tools to model and interpret experimentally observed plasma physics phenomena

Control & Automation

- **Data Acquisition Systems**: Experienced in LabVIEW and Labwindows for real-time diagnostic control, automation of multi-instrument setups, and high-speed data acquisition in experimental campaigns.

Design & Simulation

- **CAD & Engineering Design**: Proficient in SolidWorks and CATIA for mechanical design of experimental systems and diagnostics Hands-on experience operating lathe and milling (MIL) machines **for** fabrication and preparation of experimental materials and components**.**
- **Circuit Design & Simulation**: Skilled in KiCad for PCB layout and LTSpice for circuit simulation, including custom amplifier/filter design for plasma signal conditioning.
- **Electromagnetic Modeling**: Experience with Poisson Superfish for simulating electromagnetic fields and optimizing plasma source configurations.

Education

- **Ph.D. in Plasma Physics (2020)**

Fundamental plasma experimental division, Institute for Plasma Research, Gandhinagar, India.
Thesis Title: Effect of inhomogeneous magnetic field on helicon antenna produced expanding plasma.
Thesis Supervisor: Prof. Prabal K Chattopadhyay

- **Master of Science in Physics** (2012).
Devi Ahilya Vishwavidyalaya (DAVV), Indore, Madhya Pradesh, India.

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Major: Material Science
Project title: Preparation and Characterization of Graphene using Raman spectrometer.
Project leader: Prof. Vasant Sathe

- **Bachelor of Science in Computer Science** (2010).
Vikram University Ujjain, Madhya Pradesh, India
Major: Computer science and Physics

Publications

1. Guide-field-dependent electron acceleration during electron-only magnetic reconnection in the PHAse Space MApping (PHASMA) experiment, R. Nirwan, Sonu Yadav; K. J. Stevenson, G. E. Bartolo,K. Kumar, P. A. Cassak, Earl Scime, *Phys. Plasmas* 32, 122115 (2025)
2. Designing an energy analyzer for measuring non-thermal electron energy distribution functions in the PHAse space MApping (PHASMA) experiment; R. Nirwan, Sonu Yadav; T. E. Steinberger, Earl Scime; *Rev. Sci. Instrum.* 96, 123501 (2025)
3. Fast photodiode arrays for high frequency fluctuation measurements of reconnecting flux ropes, Thomas Rod, **Sonu Yadav**, Gustavo Bartolo, and Earl Scime Rev. Sci. Instrum. 95, 093525 (2024)
4. Coherent Mode and Turbulence Measurements with a Fast Camera, Gustavo Bartolo, **Sonu Yadav,** Chloelle Fitz, and Earl Scime, Rev. Sci. Instrum. 95, 093511 (2024)
5. Effect of inhomogeneous magnetic field on plasma generation in a low magnetic field helicon discharge", **Sonu Yadav**, K. K. Barada, S. Ghosh, J. Ghosh and, P. K. Chattopadhyay, Phys. Plasmas, **26**, 082109 (2019).

6. Role of ion magnetization in formation of radial density profile in magnetically expanding plasma produced by helicon antenna", **Sonu Yadav**, S. Ghosh, S. Bose, K. K. Barada, R. Pal and P. K. Chattopadhyay, Physics of Plasma, **25**, 043518 (2018).

7. Formation of annular plasma downstream by magnetic aperture in the helicon experimental" S. Ghosh, **Sonu Yadav**, K. K. Barada, P. K. Chattopadhyay, J. Ghosh, R. Pal and D. Bora, Physics of Plasmas **24**, 020703 (2017).

Submitted for review: Lower-Hybrid Drift Wave–Driven Electron Energization in Guide-Field Reconnection, Physical Review Letters, 2026

Under Preparation:

1. Suppression of Magnetic Reconnection Current Sheets in Partially Ionized Background Plasmas **Sonu Yadav** and Earl Scime; in preparation for submission (2026).
2. Characteristics of lower hybrid waves during electron-only magnetic reconnection, **Sonu Yadav** et. al. in preparation for submission (2026).
3. Tomographic Studies of the 3D Evolution of Merging Flux Ropes; Rood, T., G.E. Bartolo, E. Scime, and **Sonu Yadav**, in preparation for submission to Phys. Plasmas (2026)
4. "Demonstration of Electron Acceleration During RF Wave Damping in a Helicon Source" in preparation for submission to Phys. Plasmas (2026), Lord, J., E. Scime, **Sonu Yadav,** P. Shi
5. "A compact motorized probe design for plasma experiments," **Scime, E., W. P. Weiss, D. Curtis, Sonu Yadav**, in preparation for submission to Rev. Sci. Instrum. (2026)
6. "Electron Heating During Plasma Compression by Magnetosonic Waves" Pantor, M., C. Fowler, J. Lord, **Sonu Yadav**, and E. Scime in preparation for submission to Phys. Plasmas (2026).
7. Ion heating during electron-only magnetic reconnection in presence of strong guide field, K. J. Stevenson, Sonu Yadav, Earl Scime, in preparation for submission to Physical review letter (2026).

Selected Conferences/workshops Attended

National/International Participation:

1 Electron Trapping by Lower Hybrid Drift Waves During Guide-Field Magnetic Reconnection, 67th Annual Meeting of the APS Division of Plasma Physics,November 17-21, 2025 Long Beach, California USA

2 “Direct observation of Wave Particle interaction in PHASMA Experiments” AGU24 913 December, Washington, D.C. USA

3 2nd Solar Wind Machine Design Workshop, Dec 15-16, 2024, College Park, MD, USA

4 1st Solar Wind Machine Design Workshop, April 18-20, 2024, Los Angeles, CA, USA

5 Direct observation of Wave Particle interaction in PHASMA Experiments” 66th Annual Meeting of the APS Division of Plasma Physics, October 7-11, 2024; Atlanta, Georgia, USA

6 Generation of super thermal electrons and electron heating in guide magnetic reconnection in PHASMA, 4th MagNetUS meeting, UCLA, April 14–17, 2024; Lake Arrowhead, California, USA

7 “Plasma Fluctuation During Guide Field Electron-Only Reconnection In PHASMA” 64th Annual of APS Division of Plasma Physics, October 30–November 3, 2023; Denver, Colorado, USA

8 “Plasma fluctuation during magnetic reconnection and PHASMA Update” 3rd MagNetUS meeting, University of Auburn, June 12–15, 2022; Auburn, Alabama, USA

9 “Evolution of two flux ropes during magnetic reconnection in PHASMA” 64th Annual of APS Division of Plasma Physics, October 17–21, 2022; Spokane, Washington, USA

10 “Formation of hollow density profile in Magnetically expanding plasma produced by helicon antenna” Sonu Yadav et al. 60th annual meeting of APS Division of Plasma Physics, 5-9 Nov, 2018, Portland, Oregon, USA

11 “Effect of non-uniform magnetic field on antenna-plasma coupling efficiency in a low magnetic field helicon discharge” 71st Annual Gaseous Electronic Conference (GEC), Sonu Yadav et al. 5-9 Nov, 2018, Portland, Oregon, USA

12 “Stochastic heating in magnetized plasma produce by helical antenna” Sonu Yadav et al. 44th EPS Conference on Plasma Physics, 26-30 June, 2017, Queen's University, Belfast, Northern Ireland.

13 “Generation of off-axially localized tail electrons in helical antenna produce cylindrical plasma” 69th Annual Gaseous Electronic Conference (GEC), Sonu Yadav et al. 10-14 Oct, 2016, Ruhr University Bochum, Germany.

14 “Controllable Location of Polarization Reversal in Nonuniform Helicon Plasma” 10th Asia Plasma and Fusion Association (APFA), Sonu Yadav et al. 14-18, December, 2015, Institute for plasma research, Gandhinagar, India

15 Measurement of Electron Energy Distribution in Presence and Absence of Current Free Double Layer in Helicon Plasma” Sonu Yadav et al. 32th National Symposium on Plasma Science & Technology 07-10 November, 2017, Institute for Plasma Research, Gandhinagar, India.

16 Polarization Reversal of Helicon Wave in Nonuniform Plasma” Sonu Yadav et al. 29th National Symposium on Plasma Science & Technology and the International Conference, 8-11 December, 2014, Mahatma Gandhi University, Kottayam, Kerala, India.

Other Research Experience

- **2013** Gas Breakdown in cold cathode DC discharge
  PhD Course Work Project, 2013, Institute for Plasma Research, India. Thesis supervisor: Prof. Prabal K Chattopadhyay

- **2012** Preparation and Characterization of Graphene using Raman spectrometer
  Master thesis, 2012, Devi Ahilya Vishwavidyalaya (DAVV), Indore, India

Professional Memberships

- Member, American Physical Society (Since 2016)
- Member, European Physical Society (Since 2017)
- Member, Plasma Science Society of India (Since 2013)

Hobbies/extracurricular

Volleyball, Soccer, badminton, cricket, reading

References:

Prof. Earl Scime
(Current Post doc Supervisor)
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West Virginia University,
Morgantown, WV, USA
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Prof. Prabal K Chattopadhyay
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Gandhinagar, India
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Prof. Paul Cassak
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Dept. Physics and Astronomy
West Virginia University,
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Prof. Joydeep Ghosh
(Collaborator)
Institute for Plasma Research,
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